CYBERSECURITY AT NETWORKWALKS | WEEK 01 | PROJECT MODULE 1

Cybersecurity Lab Environment Setup

VirtualBox, Kali Linux, Windows 10 victim VM and Wireshark SOC Lab 3

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TASK REFERENCE WK1-PM1 - Lab Setup (VirtualBox and Kali Linux)

DATE 30 October 2026

STATUS Completed, including optional Phase 2 and Wireshark Lab 3

1. Executive Summary

This report documents the design, configuration and verification of a virtual cybersecurity testing laboratory built for Week 1 of the Cybersecurity program at NetworkWalks (task WK1-PM1). The lab is hosted on a personal laptop using Oracle VirtualBox and provides an isolated environment for ethical hacking, network analysis and SOC practice.

All mandatory requirements were completed: a custom NAT Network on the 10.0.0.0/24 subnet, a Kali Linux 2026.3 attacker machine on the fixed address 10.0.0.2/24, working Internet access, clipboard and drag-and-drop integration, a shared host folder, and a clean baseline snapshot. The optional work was also completed: a Windows 10 victim virtual machine was installed on the same network, and the Wireshark SOC Lab 3 ("Identify the Attacker's IP") was finished with a score of 70 out of 100.

The main technical issue encountered was loss of Internet connectivity ("Network is unreachable") after the static IP was applied. It was resolved with a NetworkManager configuration change, described in Section 6.

2. Objectives and Requirement Compliance

The table below maps each requirement of the task brief to its outcome.

Requirement	Status	Evidence
Use VirtualBox as the base (latest recommended version)	Completed	Figures 1, 2
Set up Kali Linux as the attacking machine	Completed	Figure 2
Network in subnet 10.0.0.0/24 using a NAT Network	Completed	Figure 1
Clipboard and file drag-and-drop enabled	Completed	Kali VM settings
Shared folder from host (Downloads)	Completed	Kali VM settings
Kali IP address 10.0.0.2/24	Completed	Figure 3
Kali has full Internet access	Completed	Figure 6 (fix applied)
Snapshot of the VM taken	Completed	VirtualBox snapshot
Optional: Windows 10 victim VM	Completed	Figures 4, 5
Optional: Wireshark SOC Lab 3	Completed (70/100)	Figure 7

3. Lab Architecture and Configuration

Both machines are attached to the same VirtualBox NAT Network. A NAT Network, unlike plain NAT, lets virtual machines communicate with each other while still reaching the Internet through address translation. This is the property needed for an attacker and victim lab.

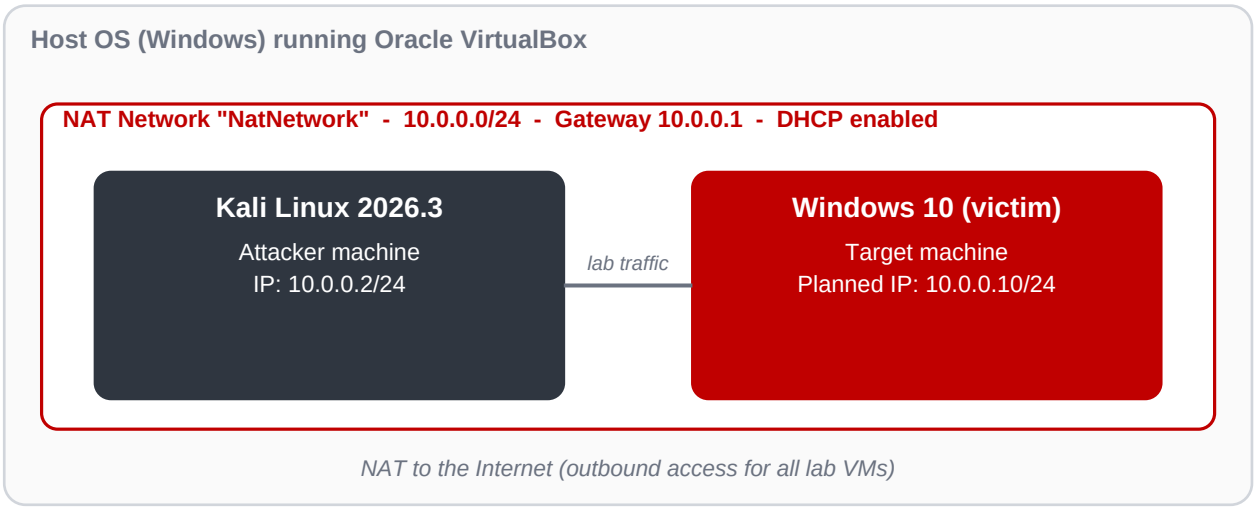


Figure: logical lab topology

Component	Configuration
Host operating system	Windows
Hypervisor	Oracle VirtualBox 7.x
Attacker VM	Kali Linux 2026.3 (VirtualBox image, amd64)
Victim VM	Windows 10 (x64), VM name "window 10 victim"
Virtual network	NAT Network "NatNetwork", 10.0.0.0/24, DHCP enabled, IPv6 disabled
Kali network settings	IP 10.0.0.2/24, gateway 10.0.0.1, DNS 8.8.8.8
Adapter	Intel PRO/1000 MT Desktop (82540EM), promiscuous mode: Allow All
Integration	Shared clipboard and drag-and-drop; shared folder (host Downloads)

4. Setup Procedure

Step 1 and 2: Install 7-Zip and VirtualBox

7-Zip was installed to handle compressed downloads, and the latest recommended version of VirtualBox was installed from the official website.

Step 3: Create the NAT Network

In VirtualBox, a NAT Network named **NatNetwork** was created under File > Tools > Network > NAT Networks with the IPv4 prefix **10.0.0.0/24**, DHCP enabled and IPv6 disabled.

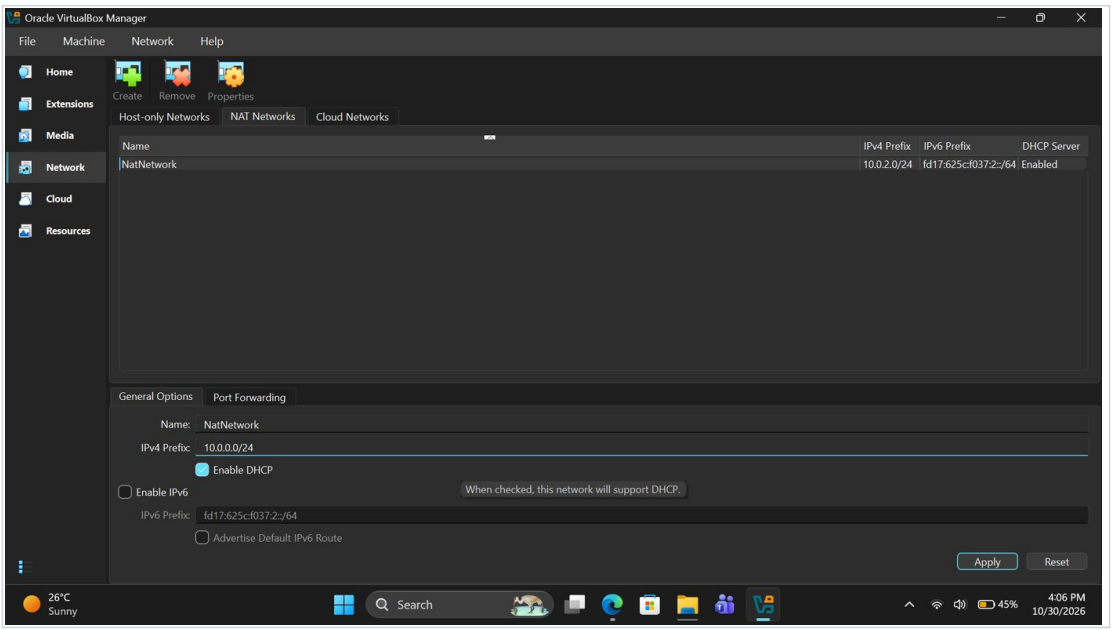


Figure 1: NAT Network configuration (10.0.0.0/24, DHCP enabled)

Step 4: Import Kali Linux

The official Kali Linux 2026.3 VirtualBox image was downloaded from kali.org and imported. Adapter 1 was attached to the NAT Network "NatNetwork", the shared clipboard and drag-and-drop were set to bidirectional, and the host Downloads folder was added as an auto-mounted shared folder.

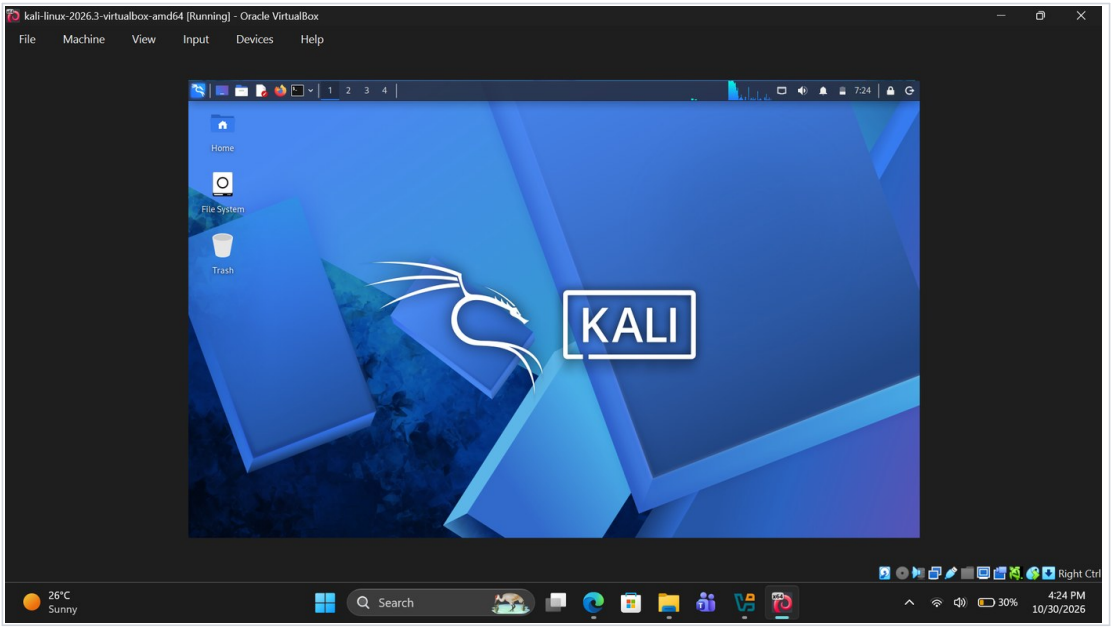


Figure 2: Kali Linux 2026.3 running in VirtualBox

Step 5: Configure the Kali IP address

A manual IPv4 configuration was applied to "Wired connection 1" in NetworkManager:

```
Method:   Manual
Address:  10.0.0.2
Netmask:  24
Gateway:  10.0.0.1
DNS:      8.8.8.8
```

The result was verified with `ip a`. Interface eth0 is UP with 10.0.0.2/24 and broadcast address 10.0.0.255.

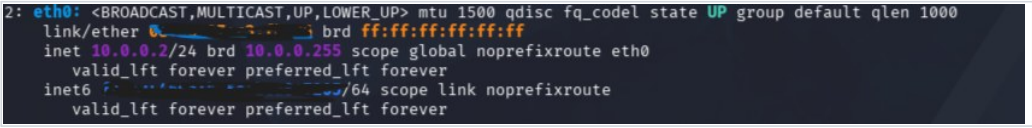


Figure 3: ip a output confirming 10.0.0.2/24 on eth0

Step 6: Baseline snapshot

Once networking worked, a snapshot of the Kali VM was taken. It serves as a known-good restore point before any scanning or exploitation exercises.

5. Optional Phase 2: Windows 10 Victim Machine

To provide a realistic target for later labs, a second VM named "window 10 victim" was created in VirtualBox and Windows 10 (x64) was installed. The VM is attached to the same NAT Network so that it can be reached from Kali. In line with the lab diagram in the task brief, the planned address for this machine is 10.0.0.10/24.

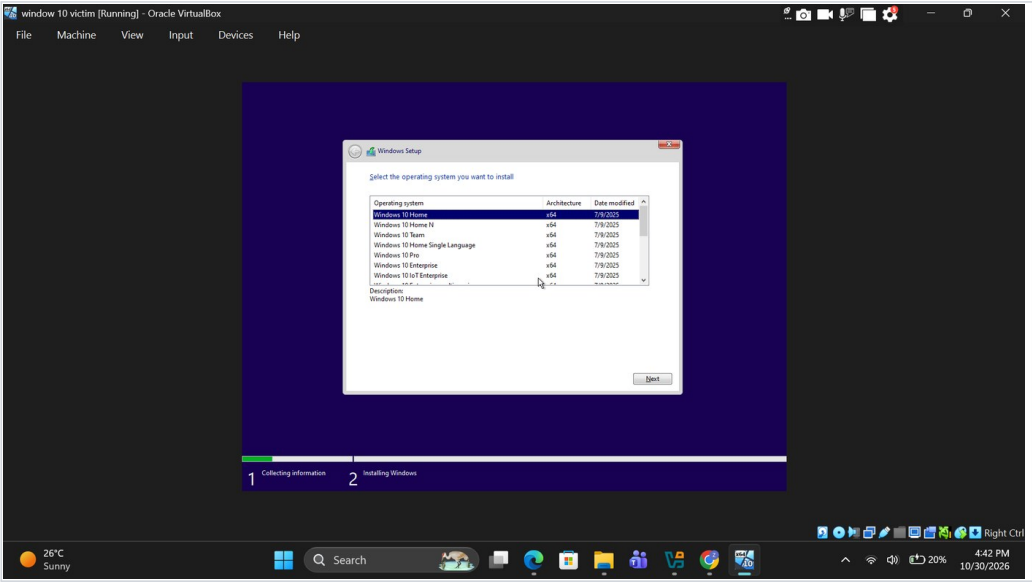


Figure 4: Selecting the Windows 10 edition during setup

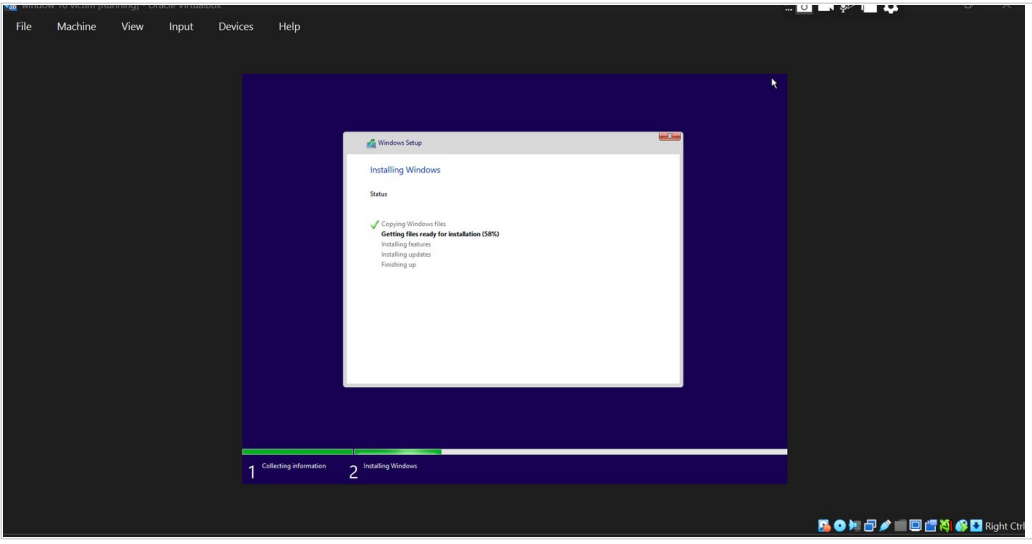


Figure 5: Windows 10 installation in progress on the victim VM

6. Problem Encountered and Resolution

Symptom. After the static address was applied, Kali could not reach the Internet. A ping to 8.8.8.8 returned "Network is unreachable".

Cause and context. The course material identifies this as a common issue with VirtualBox 7 and Kali Linux 2026.1 or newer when a manual IPv4 address is used. The recommended fix removes the duplicate address detection (DAD) delay on the NetworkManager connection.

Resolution. The following commands were run, and the connection was reactivated successfully:

```
sudo nmcli connection modify "Wired connection 1" ipv4.dad-timeout 0
sudo nmcli connection down "Wired connection 1"
sudo nmcli connection up "Wired connection 1"
```

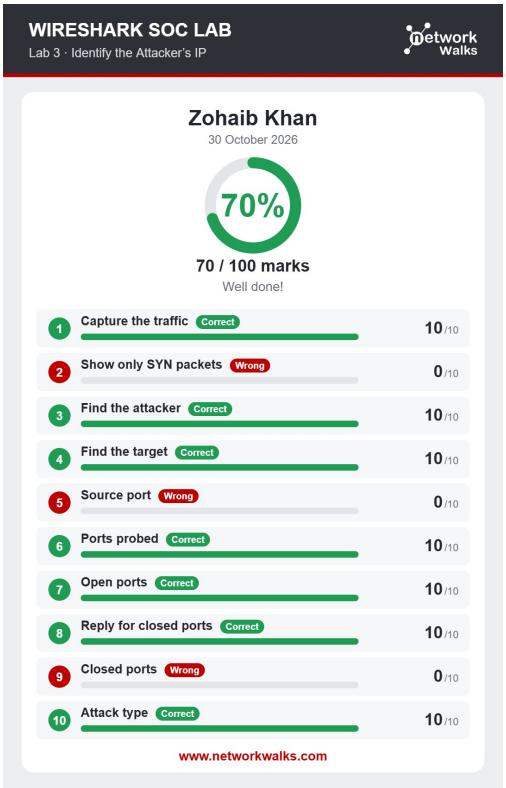


Figure 6: Failed ping, then the nmcli fix and successful reactivation of the connection

Checks used while troubleshooting:

- NAT Network created correctly with prefix 10.0.0.0/24.
- Adapter 1 attached to the correct NAT Network.
- No other VM on the network using 10.0.0.2.
- Gateway set to 10.0.0.1 and DNS to 8.8.8.8.
- Connection name matches the real NetworkManager profile name.

7. Optional: Wireshark SOC Lab 3



The optional Wireshark SOC Lab 3 simulates a SOC analyst task: examine a packet capture, then identify the attacker, the target and the type of attack.

Result: 70 / 100. Seven of ten tasks were answered correctly, covering capturing traffic, identifying the attacker and target, the ports probed, open ports, the reply for closed ports and the attack type.

Finding. The traffic is a TCP SYN (half-open) port scan, a reconnaissance technique. The scanner sends a SYN packet to many ports. An open port answers with SYN, ACK, while a closed port answers with RST, ACK. The handshake is never completed, which makes the scan quieter than a full connect scan.

Figure 7: Lab 3 result, "Identify the Attacker's IP"

Areas for improvement

Three tasks were missed: showing only SYN packets, the source port, and the closed ports. Review of these points led to the following notes:

- **SYN-only display filter.** Filtering on the SYN flag alone also matches SYN, ACK replies. A filter for initial SYN packets only is `tcp.flags.syn == 1 && tcp.flags.ack == 0`.
- **Source port.** A scanner uses a random high (ephemeral) source port; the probed service port is the destination port. The two columns must be read carefully.
- **Closed ports.** Closed ports are identified by RST, ACK replies, for example with `tcp.flags.reset == 1`, reading the source port of each reset.

8. Verification Summary

Test	Command	Expected result
IP address	ip a	eth0 shows 10.0.0.2/24 (confirmed, Figure 3)
Gateway reachable	ping -c 4 10.0.0.1	Replies received
Internet access	ping -c 4 8.8.8.8	Replies received after nmcli fix
DNS resolution	nslookup google.com	Domain resolves
Lab connectivity	ping -c 4 10.0.0.10	Replies from Windows 10 victim

9. Key Learnings

- **NAT versus NAT Network.** Plain NAT isolates each VM; a NAT Network allows VMs to talk to each other and still reach the Internet.
- **Static addressing.** Manual IPv4 settings in NetworkManager need the correct prefix, gateway and DNS, and some VirtualBox and Kali combinations need the DAD timeout adjustment.
- **Snapshots.** A clean snapshot should be taken before experiments, not after something breaks.
- **Reading a port scan.** SYN, SYN-ACK and RST-ACK patterns show which ports an attacker probed and which are open.
- **Documentation.** Recording errors, fixes and evidence is as important as the configuration itself.

10. Security and Ethical Use

This laboratory is intended strictly for education and authorised testing. All activity took place inside an isolated virtual network on the author's own machine. Tools and techniques practised here must only be used against systems that the user owns or has explicit written permission to test.

11. Conclusion and Next Steps

The Week 1 deliverable was completed in full. The lab provides a stable base for upcoming work: Kali is reachable at a fixed address, a Windows 10 target is ready on the same network, and a clean snapshot allows quick recovery. Next steps are to run ping tests between all machines, take a snapshot of the Windows 10 VM, add further target machines, and repeat the Wireshark lab to close the gaps identified in Section 7.

Tools and references: 7-Zip (7-zip.org), VirtualBox (virtualbox.org), Kali Linux (kali.org), Wireshark (wireshark.org), NetworkWalks (networkwalks.com).